
FIT5145: Foundations of Data Science
Assignment 1 (Proposal)

Name: Bharath Arun Gandhimani

Student ID: 35501308

GenAI Tool Used: ChatGPT – 5 & Google Gemini for Idea Generation and improving clarity on the report's structure.

Introduction

Formula One is often regarded as “The Pinnacle Motorsport”, where marginal improvements in performance could be the difference between winning and being out of the points. As sport becomes more digitized, F1 teams, sponsors, and supporters rely more on insights based on data to understand trends in performance, strategy, and competitiveness.

This assessment uses the Formula 1 World Championship dataset (1950–2024) provided by Kaggle and includes detailed race outcomes, drivers, constructors, lap times, pit stops, and circuit characteristics. The intention is to create tangible strategic suggestions for Max Verstappen - one of the most dominant drivers in modern F1, to improve his performance in the upcoming season.

Unlike standard statistical summaries of wins or podium finishes, the work here will take account of context-related factors such as circuit types (street, technical, high-speed), performance patterns of opponents, pit-stop effectiveness, and reliability over a season. These factors are central to determining race strategy, and data-based information can assist in optimizing driver preparation, decision-making, and general championship-winning abilities for the upcoming season.

While capitalizing on trends through history and exploratory data analysis (EDA) visually rather than through predictive regression models, this assessment focuses more on strategic insight than statistical complexity. The hope is to show how data science methods can be used to directly inform motorsport strategy, making findings applicable to different stakeholders.

Related Work

F1 data analytics is not a new thing. Teams have high expenses on telemetry, simulation, and machine learning to optimize car setup and race strategy. Past research and projects (e.g., Koh, 2024 on Kaggle) have done the following:

- Analyzing race outcomes spanning decades in order to identify top constructors and drivers.
- Examining pit stop durations and their impact on finishing places.
- Tracking driver and constructor performance trends over periods.

However, such analyses remain descriptive and retrospective, speaking to fans and excursions into data generally, as opposed to strategic deployments for a specific driver or team. Predictive race modeling has been the subject of scholarly work, as well, though such projects usually resort to complicated simulation frameworks from proprietary data sets.

This assessment stands out as being strategy and driver-specific analysis. Instead of inquiring "Who performed well in the past?", this project inquires:

- Against what competition and on what types of circuits does Max Verstappen perform most poorly?
- How do pit stop times influence his relative performance compared to competitors?
- Do seasonal consistency patterns of reliability (e.g., early-season vs. late-season races) that highlight areas of improvement exist?

This unconventional approach demonstrates novelty in three ways:

- **Driver-Centric Insight** – Most existing studies aggregate across drivers or constructors; this work isolates Max Verstappen's strategic strengths and weaknesses against specific competitors and track types.
- **Open-Data Democratization** – By deriving actionable strategy from publicly available Kaggle datasets, the project challenges the norm that only teams with proprietary telemetry can access meaningful insights.
- **Transferable Methodology** – The framework of combining context-specific factors (track category, rival patterns, pit stop dynamics, reliability trends) is underexplored in academic and fan analytics but can be extended beyond F1 to other sports or industries where open data is available but underutilized.

Business Model

Stakeholders

The primary stakeholders who benefit from this assessment are:

- **Drivers & Race Engineers** -> to refine pre-race preparation and in-race decision-making.
- **Team Strategists (Red Bull Racing)** -> to optimize pit-stop timing, tire strategy, and rival analysis.
- **F1 Management & Sponsors** -> to highlight performance narratives that can engage fans and improve brand value.
- **Fans & Media Analysts** -> to access deeper, more engaging insights about race strategy beyond surface-level statistics.

Value Proposition

This assessment aims to unlock the hidden value of publicly available Formula 1 data by providing strategic insights that are typically confined to team's proprietary telemetry systems. By categorizing circuits, benchmarking rival performance, and visualizing pit stop patterns, the project addresses several real-world challenges in motorsport:

- Strategic Decision-Making for Teams
 - The analysis highlights conditions under which drivers like Max Verstappen can gain a competitive edge, offering teams actionable intelligence without the high cost of proprietary simulation systems. For instance, knowing how rivals typically underperform on street circuits or during specific pit stop windows can inform race-day strategy and improve the probability of victory.
- Bridging the Data Gap for Sponsors and Stakeholders
 - Sponsors often lack access to the granular performance data that teams guard closely. By providing digestible insights from open data, this project allows sponsors to better align their branding with performance narratives (e.g., associating with reliability, adaptability, or dominance on specific track types), ultimately enhancing return on investment.

- Enhancing Transparency and Fan Engagement
 - Formula 1 is often criticized for being opaque, with strategy decisions hidden behind closed doors. This project democratizes access to performance analytics, making strategy more transparent and engaging for fans. Clear, visually compelling insights deepen fan understanding of why drivers win or lose, strengthening audience connection and long-term loyalty.
- Scalable Applications Beyond Motorsport
 - The methodology-using open data to uncover patterns in performance and decision-making-has broader implications for other industries where high-cost proprietary analytics create barriers. This showcases how transparent, data-driven insights can level the playing field, reduce reliance on closed systems, and drive innovation.

In doing so, this assessment not only advances competitive fairness within Formula 1 but also illustrates how open-data science can have social and commercial impact in other data-intensive fields.

References

- Kaggle (2020). Formula 1 World Championship (1950–2020) Dataset. Retrieved from: <https://www.kaggle.com/datasets/rohanrao/formula-1-world-championship-1950-2020>
- Koh, W. (2024). Formula 1 Data Analysis (2001–2024). Kaggle Notebook. Retrieved from: <https://www.kaggle.com/code/wonjoonkoh/fl-2001-2024-analysis>
- Gandomi, A. & Haider, M. (2015). Beyond the hype: Big data concepts, methods, and analytics. *International Journal of Information Management*, 35(2), 137–144.
- Verstappen, M. (2023). On Strategy and Performance in Formula 1. Red Bull Racing Media.